

Capstone Project – 01

Rule Engine Implementation

Acceptance Criteria: Rules shall vary from client to client hence the Rules have to be made configurable. The Rule Engine should be able to pick up these rules, execute them and assign an end state for the records.

Rule Execution End States

Sequence	Description	Status
1	Bank Transaction uploaded and not matched with Open AR/Remittance	Open
2	Bank Statement & Open AR/Remittance Value matched with variance	Partial Match
3	Bank Statement & Open AR/Remittance Value fully matched	Closed
4	No Invoice details in Bank Statement & No Remittance Advice	UAC (Un-applied Cash)
5	No information in Bank Statement	UIC (Un-Identified Cash)
6	User queried for more information	Query

Rules

The Rules are primarily defined on the columns from Bank Statement, Open AR and Remittance Advices configured for each deal.

Below are few Sample rules:

Priority	Bank Statement	ERP AR Open	Remittance Advice	Match	Partial Match
1	Customer Name+PO number+Amount	Customer Name+PO number (get Invoice number from it) + Amount		2 Way	Unmatched value after deduction should be posted into UAC
2	Customer Name+Delivery number+Amount	Customer Name+Delivery number (get Invoice number from it)+Amount		2 Way	Unmatched value after deduction should be posted into UAC
3	Customer Name+Invoice number+Invoice Date+Amount	Customer Name+Invoice number+Invoice Date+Amount		2 Way	Unmatched value after deduction should be posted into UAC
4	Customer Name+Invoice number+Amount	Customer Name+Invoice number+Amount		2 Way	Unmatched value after deduction should be posted into UAC
5	Customer Bank Payment	Customer Name+PO number+Invoice number+Invoice Date+Amount	Customer Name+PO number+Invoice number+Invoice Date+Amount	3Way	Unmatched value after deduction should be posted into UAC

6	Customer Bank Payment	Customer Name+PO number+Invoice number+Amount	Customer Name+PO number+Invoice number+Amount	3Way	Unmatched value after deduction should be posted into UAC
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- **Priority 1 Rule:** The Customer Name, PO Number is available, get invoice number mapped to that PO number and the Invoice Amount using the **Bank statement** where it mentioned has to be compared with **Open AR** Customer Name, Invoice Number (mapped to that PO Number) and the Invoice Amount.
- **Priority 2 Rule:** The Customer Name, Delivery Number, Invoice Number and the Invoice Amount from the **Bank statement** has to be compared with **Open AR** Customer Name, Invoice Number (mapped to delivery number) and the Invoice Amount.
- **Priority 3 Rule:** The Customer Name, Invoice Number, Invoice Date and the Invoice Amount from the **Bank statement** has to be compared with **Open AR** Customer Name, Invoice Number, Invoice Date and the Invoice Amount.
- **Priority 4 Rule:** The Customer Name, Invoice Number and the Invoice Amount from the **Bank statement** has to be compared with **Open AR** Customer Name, Invoice Number and the Invoice Amount.
- **Priority 5 Rule:** The Customer Name, PO Number, Invoice Number, Invoice Date and the Invoice Amount from the **Remittance Advice** has to be compared with **Open AR** Customer Name, PO Number, Invoice Number, Invoice Date and the Invoice Amount.
- **Priority 6 Rule:** The Customer Name, PO Number, Invoice Number and the Invoice Amount from the **Remittance Advice** has to be compared with **Open AR** Customer Name, PO Number, Invoice Number and the Invoice Amount.

Capstone Project – 02

Deduction Identification:

Identify Short Payment

Given an invoice amount exists, When the received payment amount is less than the invoice amount. Then the system should calculate the difference as a potential deduction.

Example

Invoice Amount	Payment Received	Deduction
10,000	9,500	500

Read Deduction Reason from Remittance

Given a remittance advice contains deduction remarks. When the payment is processed, then the system should capture the deduction reason.

Supported Reasons

- Pricing Issue
- Freight Claim
- Damage Claim
- Tax Difference
- Discount Taken

Create Deduction Record

Given a deduction is identified, When the analyst posts the cash application. Then the system should create a deduction/dispute item for the remaining balance.

Apply Partial Cash

Given a customer partially paid an invoice. When the payment is applied. Then the system should:

- close the paid portion
- leave the deduction amount open for investigation

Assign Deduction Reason Code

Given a deduction exists. When the analyst selects a reason. Then the system should assign a predefined deduction reason code.

Sample Reason Codes

Code	Description
D01	Pricing
D02	Freight
D03	Damage
D04	Tax
D05	Discount

Auto Match Based on Tolerance

Given the difference amount is within tolerance. When the payment is processed. Then the system may auto-write off the difference instead of creating a deduction.

Process Flow

1. Receive customer payment
2. Import remittance advice
 - 2.1 Reach out to collector/customer if not received
3. Match invoice number
4. Compare invoice vs payment amount
 - 4.1 Perform the **reconciliation** (examples below)
5. Identify deduction difference/Over Payment
6. Capture deduction reason (if available)
7. Apply payment
8. Create deduction/dispute item
9. Route deduction for resolution